UT Health Houston

Air Pollution & Asthma Burden in the United States

A County-level study of Air Quality Index (AQI) and Emergency Department Visits

PH 1830L: Categorical Data Analysis

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Motivation for Studying Air Pollution and Asthma

The Burden of Asthma

- About 23 million U.S. adults have asthma[1]
- Asthma leads to ~1.6 million emergency department visits each year[2]

Why Air Pollution Matters

- Poor air quality (AQI) reflects exposure to harmful pollutants
- Air pollution is associated with increased asthma attacks and ED visits[3]

Key Question

- Is air pollution driving asthma emergencies, or are we observing correlation rather than causation?

[1] Asthma and Allergy Foundation of America (AAFA)
[2] Centers for Disease Control and Prevention (CDC)
[3] U.S. Environmental Protection Agency (EPA)

Research Question

Is air pollution exposure, measured using both continuous (annual weighted AQI) and categorical (AQI category) indicators, associated with elevated asthma-related emergency department visit rates across U.S. counties after adjusting for state-level variation?

Hypotheses

Null Hypothesis (H_0)

$\beta_{(AQI)} = 0$; Annual weighted AQI is not associated with elevated asthma ED visit rates after adjusting for state.

Alternate Hypothesis (H_1)

$\beta_{(AQI)} \neq 0$; Annual weighted AQI is associated with elevated asthma ED visit rates after adjusting for state.

Data Sources

Asthma Data

- CDC asthma emergency department visit data
- County-level, age-adjusted ED visit rates per 10,000 (county-specific, sex-stratified)

Air Quality Data

- U.S. EPA Air Quality Index (AQI) data (county-level, 2023)
- Includes counts of days by AQI category and pollutant type

Constructed Variables

Outcome (Asthma Burden)

Asthma ED Status (Binary)

- Elevated vs Not Elevated
- Based on sex-specific national benchmarks (Pate et al., 2024)
- Derived from age-adjusted ED visit rates

Exposure (Air Pollution)

Annual Weighted AQI (Continuous)

- Weighted average using AQI category midpoints
- Reflects both frequency and severity of pollution

AQI Category (Low, Moderate, High)

- Based on proportion of unhealthy AQI days
- Categorized using tertiles (balanced groups)

Pollutant Category (Nominal)

- Dominant pollutant based on highest number of AQI days
- Categories: CO, NO_2, Ozone, PM2.5, PM10, Mixed

Merged Dataset

Dataset Overview

- County-level dataset linking air pollution exposure and asthma outcomes
- Cross-sectional data (2023)
- Each Observation represents a county **(unit of analysis)**

Table 1. Summary of the Merged County-Level Dataset (United States, 2023)				
Number of Observations (County-Level)	Number of States	Number of Counties	Sex Categories	Year
528	13	234	2	2023

Note: Data were merged by State, County, and Year; unmatched observations were dropped (final n = 528).

Primary Variables

Outcome (Response Variable)

- **Asthma ED Status (Binary)**
  Elevated vs Not Elevated (sex-specific thresholds)

Covariate

- **State**
  Accounts for regional variation (e.g., policy, environment, healthcare access)
  A Key Confounder

Exposure

- **Annual Weighted AQI (Continuous)**
- **prop_unhealthy (Continuous)**
  Proportion of days with unhealthy AQI
- **AQI Category (Low/Moderate/High)**
  Based on tertiles of prop_unhealthy
- **Pollutant Category (Nominal)**
  Dominant pollutant (CO, NO_2, Ozone, PM2.5, PM10, Mixed)

Identifiers/Supporting Variables

- **County**
- **Year** (2023)
- **Sex** (used to define outcome thresholds)

Summary Statistics **Categorical Predictors**

Table 2. Descriptive Summary of Categorical Variables (N = 528)

Characteristic	N = 528[1]
AQI Category	
Low	192 (36%)
Moderate	158 (30%)
High	178 (34%)
Pollutant Category	
Ozone	208 (39%)
PM2.5	304 (58%)
PM10	16 (3.0%)
Sex	
Female	264 (50%)
Male	264 (50%)
Asthma ED Status	
Not Elevated	224 (42%)
Elevated	304 (58%)

[1] n (%)

Air Pollution Exposure

- AQI categories, defined using the proportion of unhealthy days, are fairly balanced across counties
  Low (36%) | Moderate (30%) | High: 34%
- Suggests good variability for analysis

Pollutant Distribution

- PM2.5 is dominant (58%)
- Ozone is substantial (39%)
- PM10 is rare (3%)
- Indicates **fine particulate matter** is the primary exposure

Population Structure

- Equal representation by sex (50/50)

Asthma Burden

- 58% of counties have elevated asthma ED rates
- Suggests substantial asthma burden nationwide

Summary Statistics **Continuous Predictors**

Table 3. Summary of Continuous Variables (N = 528)

Characteristic	N = 528
Proportion Unhealthy Days	
Mean (SD)	0.02 (0.02)
Median	0.01
Min – Max	0.00 – 0.19
Asthma ED Visit Rate	
Mean (SD)	35.01 (12.89)
Median	31.85
Min – Max	5.10 – 82.10
Annual Weighted AQI	
Mean (SD)	41.56 (9.15)
Median	41.06
Min – Max	25.14 – 79.66

Air Pollution Exposure

- Average AQI: 41.6 (St. Dev. 9.2)
- Range: 25 to 80
- Indicates substantial variability in pollution level across counties

Unhealthy Air Exposure

- Mean proportion of unhealthy days: 2%
- Median: 1%
- Most counties experience **low** but **non-zero** exposure to harmful air

Asthma Burden

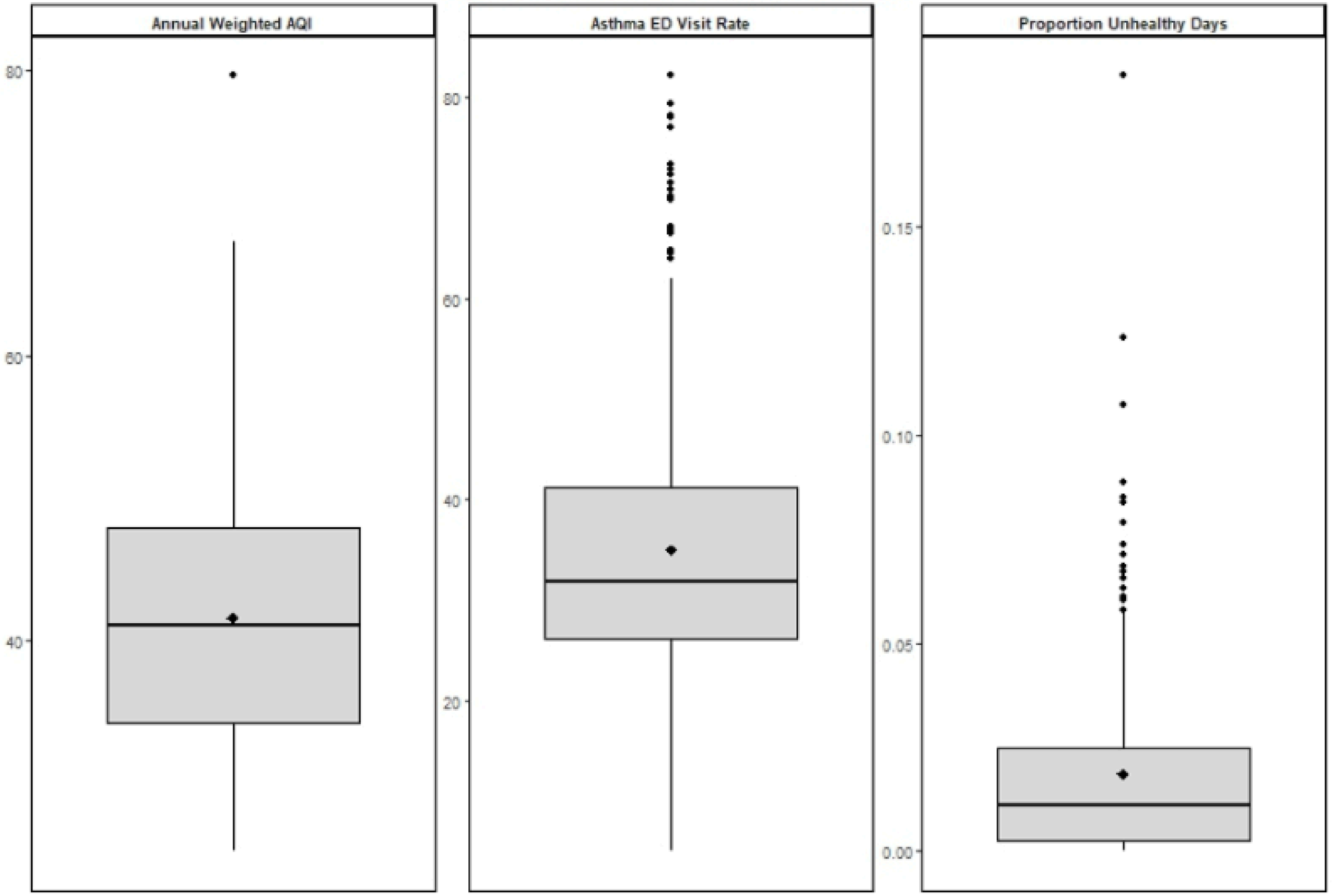
- Mean ED visit rate: 35 per 10,000
- Wide range: 5 to 82 per 10,000
- Suggests large variation in asthma burden across counties

Distribution of Continuous Variables

Key Insights

- Asthma ED visit rates show wide variability and multiple outliers
- AQI is more centrally distributed with moderate spread
- Proportion of unhealthy days is low overall but right-skewed
- Indicates **heterogeneity** in both exposure and outcomes across counties

Figure 2. Box Plots of Continuous Variables



Bivariate Analysis

AQI Category x Asthma ED Status

Table 4. Association Between AQI Category and Asthma ED Status

	Asthma_ED_Status		
AQI Category	Not Elevated	Elevated	Total
AQI_Category			
Low	60 (31%)	132 (69%)	192 (100%)
Moderate	65 (41%)	93 (59%)	158 (100%)
High	99 (56%)	79 (44%)	178 (100%)
Total	224 (42%)	304 (58%)	528 (100%)

Key Findings

- Higher AQI exposure is associated with lower proportion of elevated asthma cases
- **Low AQI counties:** 69% elevated asthma burden
- **High AQI counties:** 44% elevated asthma burden
- Indicates a negative association between AQI category and asthma burden
- Suggests a **counterintuitive pattern given prior assumptions**

Possible Explanations

- Confounding by state-level factors (healthcare, policy, demographics)
- Differences in healthcare access and utilization
- Urban vs rural variation
- Ecological (county-level) analysis limitations

This highlights the importance of adjusting for confounders, which is addressed in the regression analysis

Bivariate Analysis

Association Between AQI Category and Asthma ED Status

Pearson Chi-Square Test

Table 5. Chi-Square Test of Association Between AQI Category and Asthma ED Status

Chi-Square Statistic	Degrees of Freedom	p-value
22.61	2	<0.001

- Statistically **significant** ($p < 0.001$) association between AQI category and asthma ED status
- Distribution of elevated vs non-elevated cases (asthma burden) differs across AQI categories

Likelihood Ratio (G^2) Test

Table 6. Likelihood Ratio Test (G-Test) of Association Between AQI Category and Asthma ED Status

G Statistic	Degrees of Freedom	p-value
22.74	2	<0.001

- Statistically **significant** ($p < 0.001$) association between AQI category and asthma ED status
- Distribution of asthma ED status differs across AQI categories. Results are **consistent** with the chi-square test

Bivariate Analysis

Trend Analysis Across AQI Categories and Asthma ED Status

Cochran-Armitage Trend Test

Table 7. Cochran–Armitage Trend Test Across AQI Categories and Asthma ED Status

Z Statistic	p-value
4.73	<0.001

- **Z = 4.73, p < 0.001:** Statistically **significant** ordered trend across AQI categories
- Elevated asthma burden decreases from low to high AQI levels
- Indicates a significant inverse relationship

This inverse pattern is likely driven by confounding and structural differences across counties rather than a true protective effect of air pollution.

- **High AQI** counties are often **urban areas** with better healthcare access, potentially leading to lower emergency visit rates despite higher pollution

Bivariate Analysis

Residual Analysis Across AQI Categories and Asthma ED Status

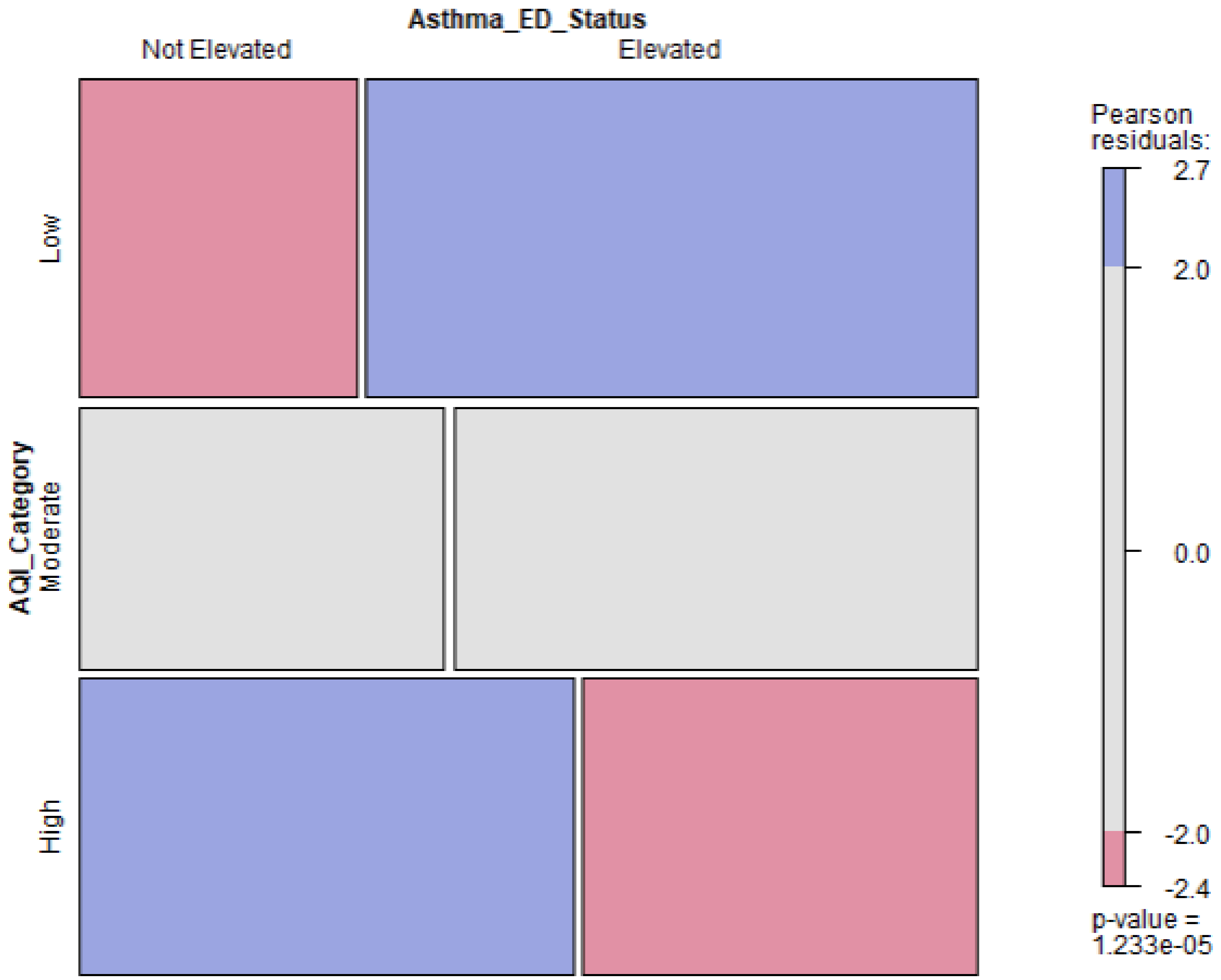
Key Findings

Blue = More than Expected

Red = Less than Expected

- Significant deviations from expected counts occur in low and high AQI categories
- Low AQI counties show more elevated asthma cases than expected
- High AQI counties show fewer elevated cases than expected
- Moderate AQI counties are close to expected levels
- Association is primarily driven by low and high AQI groups

Figure 3. Mosaic Plot of AQI Category and Asthma ED Status



Confirms the observed inverse pattern

Bivariate Analysis

Pollutant Category x Asthma ED Status

Table 8. Association Between Pollutant Category and Asthma ED Status

Pollutant Category	Asthma_ED_Status: Not Elevated	Asthma_ED_Status: Elevated	Total
Pollutant_Category			
Ozone	90 (43%)	118 (57%)	208 (100%)
PM2.5	125 (41%)	179 (59%)	304 (100%)
PM10	9 (56%)	7 (44%)	16 (100%)
Total	224 (42%)	304 (58%)	528 (100%)

Key Findings

- Counties dominated by ozone and PM2.5 have comparable proportions of elevated asthma cases
- PM10 counties are few and show no clear pattern
- Overall, no consistent differences in asthma burden by dominant pollutant

Possible Explanations

- Overall pollution burden may matter more than pollutant type
- Dominant pollutant may oversimplify true exposure
- County-level analysis may mask within-county variation

This highlights that overall pollution exposure may matter more than pollutant type

Bivariate Analysis

Association Between Pollutant Category and Asthma ED Status

Pearson Chi-Square Test

- No statistically **significant** (p > 0.001) association between Pollutant category and asthma ED status
- Distribution of elevated vs non-elevated cases (asthma burden) is similar across pollutant types

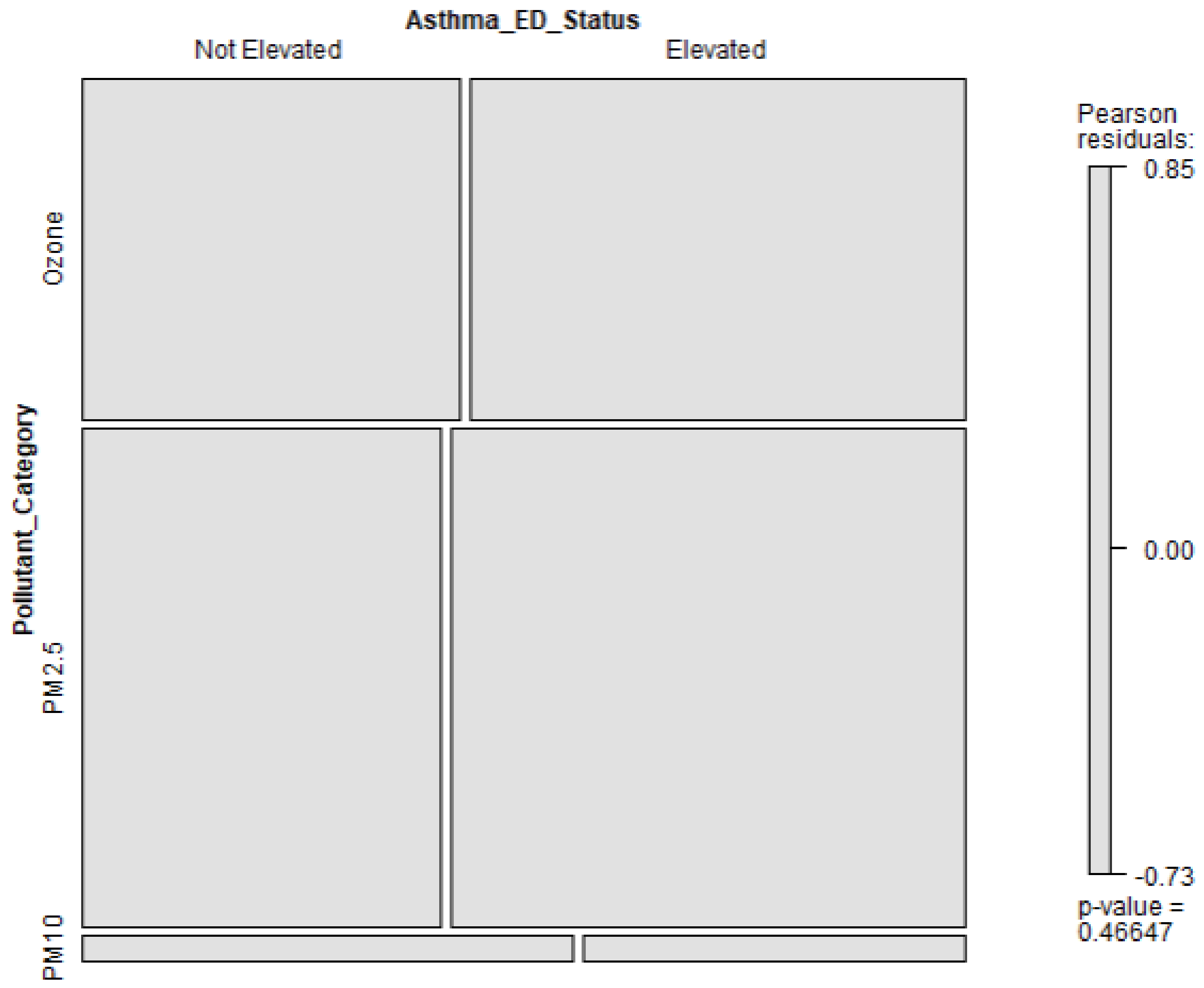
Residual Analysis Between Pollutant Category and Asthma ED Status

- Standardized residuals are small across all categories (within ±2)
- No cells show meaningful deviation from expected counts
- No single pollutant category drives the association
- Findings are **consistent** with non-significant test results

Table 9. Chi-Square Test of Association Between Pollutant Category and Asthma ED Status

Chi-Square Statistic	Degrees of Freedom	p-value
1.53	2	0.466

Figure 4. Mosaic Plot of Pollutant Category and Asthma ED Status



Stratified Analysis By Sex

Table 10. AQI Category and Asthma ED Status, Stratified by Sex

AQI_Category	Not Elevated	Elevated
Female		
Low	30	66
Moderate	35	44
High	54	35
Male		
Low	30	66
Moderate	30	49
High	45	44

Key Findings

- In both females and males, elevated asthma burden is highest in low AQI counties
- The proportion of elevated cases decreases as AQI increases
- This Inverse pattern is consistent across sexes after applying sex-specific thresholds
- Suggests the association is **not** explained by sex differences

Cochran–Mantel–Haenszel Test (Adjusted for Sex)

Table 11. Cochran–Mantel–Haenszel Test of Association Between AQI Category and Asthma ED Status, Adjusted for Sex

CMH Chi-Square	Degrees of Freedom	p-value
22.59	2	<0.001

Key Findings

- CMH $\chi^2 = 22.59$, df = 2, $p < 0.001$
- Association remains statistically significant after adjusting for sex
- Elevated asthma burden decreases as AQI increases in both sexes

The relationship between AQI and asthma burden is robust and not explained by sex

Stratified Analysis By State

Cochran–Mantel–Haenszel Test (Adjusted for State)

Table 12. Cochran–Mantel–Haenszel Test Adjusted for State

CMH Chi-Square	Degrees of Freedom	p-value
1.84	2	0.398

Key Findings

- CMH χ^2 = 1.84, df = 2, p =0.398
- No statistically significant association **after** adjusting for state
- AQI–asthma relationship disappears after accounting for state-level variation
- Previously observed AQI–asthma relationship largely explained by differences across states

Assessment of State Level Confounding

Key Findings

Crude Model (Unadjusted):

- **OR 0.36:** High AQI associated with lower odds of elevated asthma burden
- Statistically significant (p<0.001)

Adjusted Model (Controlling for State):

- Association between AQI and asthma no longer significant
- Effect estimates move toward null (OR ≈ 1)
- Indicates strong confounding by state

Table 13. Odds Ratios for Asthma ED Status by AQI Category (Crude and Adjusted for State)

Model	OR (95% CI)	p-value
High vs Low		
Adjusted	1.41 (0.72, 2.78)	0.311
Crude	0.36 (0.24, 0.55)	<0.001
Moderate vs Low		
Adjusted	1.53 (0.83, 2.82)	0.173
Crude	0.65 (0.42, 1.01)	0.055

The observed AQI–asthma association is confounded by state-level factors

Simple Logistic Regression

Logistic Regression (Continuous AQI)

Predictors: 1 (Annual Weighted AQI)

Table 14. Logistic Regression Results: Continuous AQI Model

Term	Coefficient	Std. Error	z value	p-value	OR	2.5% (OR)	97.5% (OR)
(Intercept)	0.5901	0.4104	1.4378	0.150	1.8042	0.8085	4.0520
Annual_Weighted_AQI	-0.0068	0.0096	-0.7110	0.477	0.9932	0.9746	1.0121

Key Findings

- OR ≈ 0.99, $p = 0.477$
- No statistically significant association between Annual Weighted AQI and asthma ED status across U.S. counties
- 95% CI includes 1 → **no evidence of effect**

Simple Logistic Regression

Logistic Regression: AQI Category (Unadjusted)

Predictors: 1 (AQI Category)

Table 15. Logistic Regression Results: AQI Category (Unadjusted Model)

Term	Coefficient	Std. Error	z value	p-value	2.5% (log-odds)	97.5% (log-odds)	OR	2.5% (OR)	97.5% (OR)
(Intercept)	0.7885	0.1557	5.0640	<0.001	0.4885	1.1001	2.2000	1.6299	3.0044
AQI_CategoryModerate	-0.4302	0.2245	-1.9169	0.055	-0.8720	0.0090	0.6503	0.4181	1.0090
AQI_CategoryHigh	-1.0141	0.2168	-4.6777	<0.001	-1.4433	-0.5926	0.3627	0.2362	0.5529

Key Findings

- AQI category is associated with asthma ED status across U.S. counties
- Moderate vs Low:
  OR ≈ 0.65
  $p = 0.055$ (not statistically significant)
- High vs Low:
  OR ≈ 0.36
  $p < 0.001$ (statistically significant)

- Higher AQI categories are associated with lower odds of elevated asthma burden (inverse relationship)

The apparent association in the categorical model likely reflects non-linear patterns and underlying geographic differences, rather than a true independent effect of AQI.

Evaluating the Contribution of AQI Category to Model Fit

Likelihood Ratio Test

Compared

- Null Model (No Predictors)
- Model with AQI Category

Table 16. Likelihood Ratio Test: Contribution of AQI Category

Comparison	Chi-Square	df	p-value
AQI Category vs Null Model	22.74	2	<0.001

INSIGHT: Adding AQI category significantly improves model fit

- Indicates AQI category is associated with asthma ED status in the unadjusted model

Multiple Logistic Regression Models

Base Model

Annual Weighted AQI + State

Expanded Model

Annual Weighted AQI + State + Pollutant Category

Table 17. Full Logistic Regression Output: Base vs Expanded Model

Term	Base_Coefficient	Base_SE	Base_z	Base_p	Base_LCI_log	Base_UCI_log	Base_OR	Base_LCI_OR	Base_UCI_OR	Expanded_Coefficient	Expanded_SE	Expanded_z	Expanded_p	Expanded_LCI_log	Expanded_UCI_log	Expanded_OR	Expanded_LCI_OR	Expanded_UCI_OR
(Intercept)	-1.2418	0.6848	-1.8135	0.07	-2.6065	0.0910	0.2889	0.0738	1.0953	-0.6912	0.8228	-0.8400	0.401	-2.3153	0.9270	0.5010	0.0987	2.5269
Annual_Weighted_AQI	0.0279	0.0126	2.2173	0.027	0.0035	0.0529	1.0283	1.0035	1.0543	0.0170	0.0154	1.1005	0.271	-0.0132	0.0476	1.0171	0.9869	1.0487
StateFlorida	4.5133	1.0873	4.1508	<0.001	2.7630	7.4642	91.2203	15.8473	1744.3817	4.2395	1.1098	3.8200	<0.001	2.4250	7.2137	69.3733	11.3027	1357.9678
StateIndiana	-0.3897	0.4745	-0.8213	0.412	-1.3271	0.5454	0.6773	0.2652	1.7252	-0.6193	0.5125	-1.2084	0.227	-1.6354	0.3880	0.5383	0.1949	1.4741
StateIowa	-0.6526	0.5484	-1.1900	0.234	-1.7463	0.4159	0.5207	0.1744	1.5158	-0.8897	0.5818	-1.5293	0.126	-2.0492	0.2438	0.4108	0.1288	1.2761
StateKansas	0.5977	0.6007	0.9950	0.32	-0.5686	1.8044	1.8179	0.5663	6.0761	0.4407	0.6190	0.7121	0.476	-0.7630	1.6823	1.5539	0.4663	5.3779
StateNebraska	-0.2795	0.6332	-0.4415	0.659	-1.5494	0.9554	0.7561	0.2124	2.5997	-0.5420	0.6713	-0.8073	0.42	-1.8908	0.7636	0.5816	0.1510	2.1460
StateNew Jersey	1.9085	0.6710	2.8444	0.004	0.6609	3.3385	6.7430	1.9365	28.1757	1.6427	0.7052	2.3293	0.02	0.3188	3.1289	5.1690	1.3754	22.8481
StateNew York	1.7073	0.5431	3.1434	0.002	0.6611	2.8045	5.5140	1.9369	16.5193	1.4458	0.5875	2.4610	0.014	0.3041	2.6233	4.2453	1.3553	13.7816
StateNorth Carolina	0.2796	0.4728	0.5914	0.554	-0.6488	1.2175	1.3226	0.5227	3.3786	-0.0104	0.5304	-0.0196	0.984	-1.0584	1.0361	0.9897	0.3470	2.8183
StateOregon	-1.6995	0.5974	-2.8449	0.004	-2.9315	-0.5638	0.1828	0.0533	0.5691	-2.0949	0.6799	-3.0812	0.002	-3.4791	-0.7915	0.1231	0.0308	0.4532
StateRhode Island	1.7689	1.1697	1.5123	0.13	-0.2396	4.8005	5.8644	0.7869	121.5747	1.3425	1.2197	1.1007	0.271	-0.7871	4.4301	3.8285	0.4552	83.9375
StateSouth Carolina	2.6699	0.8389	3.1826	0.002	1.1961	4.6317	14.4383	3.3071	102.6867	2.3693	0.8748	2.7084	0.007	0.8076	4.3798	10.6902	2.2425	79.8214
StateWisconsin	-1.2820	0.5106	-2.5109	0.012	-2.3030	-0.2887	0.2775	0.1000	0.7492	-1.4939	0.5416	-2.7584	0.006	-2.5775	-0.4409	0.2245	0.0760	0.6435
Pollutant_CategoryPM2.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.2981	0.2876	1.0368	0.3	-0.2653	0.8647	1.3473	0.7670	2.3743
Pollutant_CategoryPM10	NA	NA	NA	NA	NA	NA	NA	NA	NA	-0.3990	0.6483	-0.6153	0.538	-1.7234	0.8523	0.6710	0.1785	2.3451

- AQI: OR ≈ 1.03, p = 0.026
  → Small but statistically significant positive association
- Several states significantly associated with asthma ED status
- Indicates substantial variation across states

- AQI: OR ≈ 1.02, p = 0.27
  → not statistically significant association
- **Pollutant category:** not significant
- State remains strongly associated

Adding pollutant category removes the AQI effect, while state remains the dominant predictor

Multiple Logistic Regression Models

Alternative Model

AQI Category + State

Table 17. Logistic Regression Results: AQI Category + State (Alternative Model)

Term	Coefficient	Std. Error	z value	p-value	2.5% (log-odds)	97.5% (log-odds)	OR	2.5% (OR)	97.5% (OR)
(Intercept)	-0.2370	0.4345	-0.5455	0.585	-1.1015	0.6186	0.7890	0.3324	1.8563
AQI_CategoryModerate	0.4227	0.3100	1.3633	0.173	-0.1813	1.0370	1.5260	0.8342	2.8208
AQI_CategoryHigh	0.3469	0.3425	1.0129	0.311	-0.3219	1.0241	1.4147	0.7248	2.7846
StateFlorida	4.5174	1.0926	4.1345	<0.001	2.7518	7.4736	91.5960	15.6706	1760.9715
StateIndiana	-0.5607	0.4807	-1.1663	0.244	-1.5132	0.3833	0.5708	0.2202	1.4672
StateIowa	-0.5985	0.5469	-1.0944	0.274	-1.6879	0.4685	0.5496	0.1849	1.5976
StateKansas	0.5206	0.5949	0.8752	0.382	-0.6356	1.7146	1.6831	0.5296	5.5545
StateNebraska	-0.5305	0.6284	-0.8441	0.399	-1.7948	0.6910	0.5883	0.1662	1.9956
StateNew Jersey	1.7220	0.6757	2.5487	0.011	0.4623	3.1584	5.5959	1.5877	23.5339
StateNew York	1.4456	0.5285	2.7354	0.006	0.4237	2.5101	4.2442	1.5276	12.3061
StateNorth Carolina	0.2102	0.4717	0.4457	0.656	-0.7178	1.1435	1.2340	0.4878	3.1377
StateOregon	-1.7878	0.5921	-3.0195	0.002	-3.0116	-0.6650	0.1673	0.0492	0.5143
StateRhode Island	1.4495	1.1700	1.2389	0.215	-0.5616	4.4811	4.2609	0.5703	88.3291
StateSouth Carolina	2.6961	0.8430	3.1981	0.001	1.2127	4.6636	14.8222	3.3625	106.0194
StateWisconsin	-1.3649	0.5268	-2.5907	0.01	-2.4181	-0.3406	0.2554	0.0891	0.7113

- AQI Category (vs Low):
  Moderate: OR ≈ 1.52, p = 0.17 → not significant
  High: OR ≈ 1.41, p = 0.31 → not significant
- **State effects:** Several states remain statistically significant
- Indicates strong variation across states

OBSERVATION

Unadjusted:

- **AQI Category** appears statistically significant
- **Annual Weighted AQI** appears not statistically significant

Adjusted for State:

- **AQI Category** appears statistically not significant
- **Annual Weighted AQI** becomes statistically significant

Possible Explanation:

- **AQI Category** reflects between-state differences (confounding)

Counties in certain AQI categories are concentrated in specific states, so the association partly reflects state-level differences rather than AQI itself

- **Annual Weighted AQI** captures within-state variation

After adjusting for state, AQI compares counties within the same state, revealing smaller, underlying differences in exposure

Multiple Logistic Regression Models

Model Comparison: AIC + LRT

Table 18. Model Comparison (AIC and Likelihood Ratio Tests)

Model	AIC	Comparison	ChiSq	df	p
Base (Annual Weighted AQI + State)	541.5066	—	—	—	—
Expanded (+ Pollutant)	544.0061	Base vs Expanded	1.5	2	0.472
Alternative (AQI Category + State)	546.6419	Base vs Alternative	-3.14	1	NA

AIC Comparison:

- Differences in AIC across models are small (< 5 units)
- Indicates models have similar overall fit
- Adding pollutant category does not meaningfully improve fit
- Categorizing AQI results in a worse-performing model

Likelihood Ratio Tests:

Base vs Expanded (Pollutant added):

- ΔDeviance = 1.50, $p = 0.47$ → not significant

Base vs Alternative (AQI categorical):

- No meaningful improvement in fit

Key Findings

The Base Model provides the best fit; additional variables do not improve model performance

Multiple Logistic Regression Models

Interaction Effects (Effect Modification Testing)

Table 19. Likelihood Ratio Tests for Interaction Effects

Interaction	Chi-Square	df	p-value
AQI Category × State	16.87	19	0.599
Annual Weighted AQI × State	19.45	12	0.078
Annual Weighted AQI × Pollutant Category	1.30	2	0.521

Annual Weighted AQI & State:

- $p = 0.078$ → not statistically significant
- Weak evidence, but insufficient to support interaction

AQI Category & State:

- $p = 0.60$ → not significant
- No variation across states

Annual Weighted AQI & Pollutant:

- $p = 0.52$ → not significant
- No variation by pollutant type

No evidence that AQI effects differ across states or pollutant types

State influences the level of asthma (confounding), not the effect of AQI (no interaction)

Multiple Logistic Regression Models

Variable Selection

Table 19. Variable Selection Results Using Stepwise Procedures

Selection Method	Final Model	AIC
Forward Selection	Asthma_ED_Status ~ State + Annual_Weighted_AQI	541.51
Backward Selection	Asthma_ED_Status ~ Annual_Weighted_AQI + State	541.51
Stepwise Selection	Asthma_ED_Status ~ Annual_Weighted_AQI + State	541.51

Forward Selection:
- Selected State first, then AQI
- Pollutant category not retained

Backward Selection:
- Removed pollutant category
- Retained State + AQI

Stepwise Selection:
- Same final model: State + AQI
- Pollutant category did not improve fit

All methods selected the same model: **State + Annual Weighted AQI**

Multiple Logistic Regression Models

Assessment of Multicollinearity

Table 20. Variance Inflation Factors (VIF) for Final and Expanded Models

Variable	VIF (Final Model)	Df.x	GVIF^(1/(2*Df)).x	VIF (Expanded Model)	Df.y	GVIF^(1/(2*Df)).y
Annual_Weighted_AQI	1.21	1	1.10	1.83	1	1.35
State	1.21	12	1.01	1.90	12	1.03
Pollutant_Category	NA	NA	NA	2.10	2	1.20

Assessment:

- All predictors have low adjusted GVIF values (~1.0–1.35)
- No variable exceeds common thresholds for concern

LOW VIF Indicates:

- Predictors provide independent information
- Model estimates are stable

Common VIF/GVIF Thresholds:

- ≈ 1 → No correlation (ideal)
- < 5 → Acceptable (no concern)
- 5 – 10 → Moderate multicollinearity
- > 10 → High multicollinearity (problematic)

No evidence of multicollinearity in either model

Multiple Logistic Regression Models

Model Diagnostics

Upper Band: counties with elevated asthma

Lower Band: counties with not elevated asthma

Figure 5: Pearson Residuals vs Observation Index

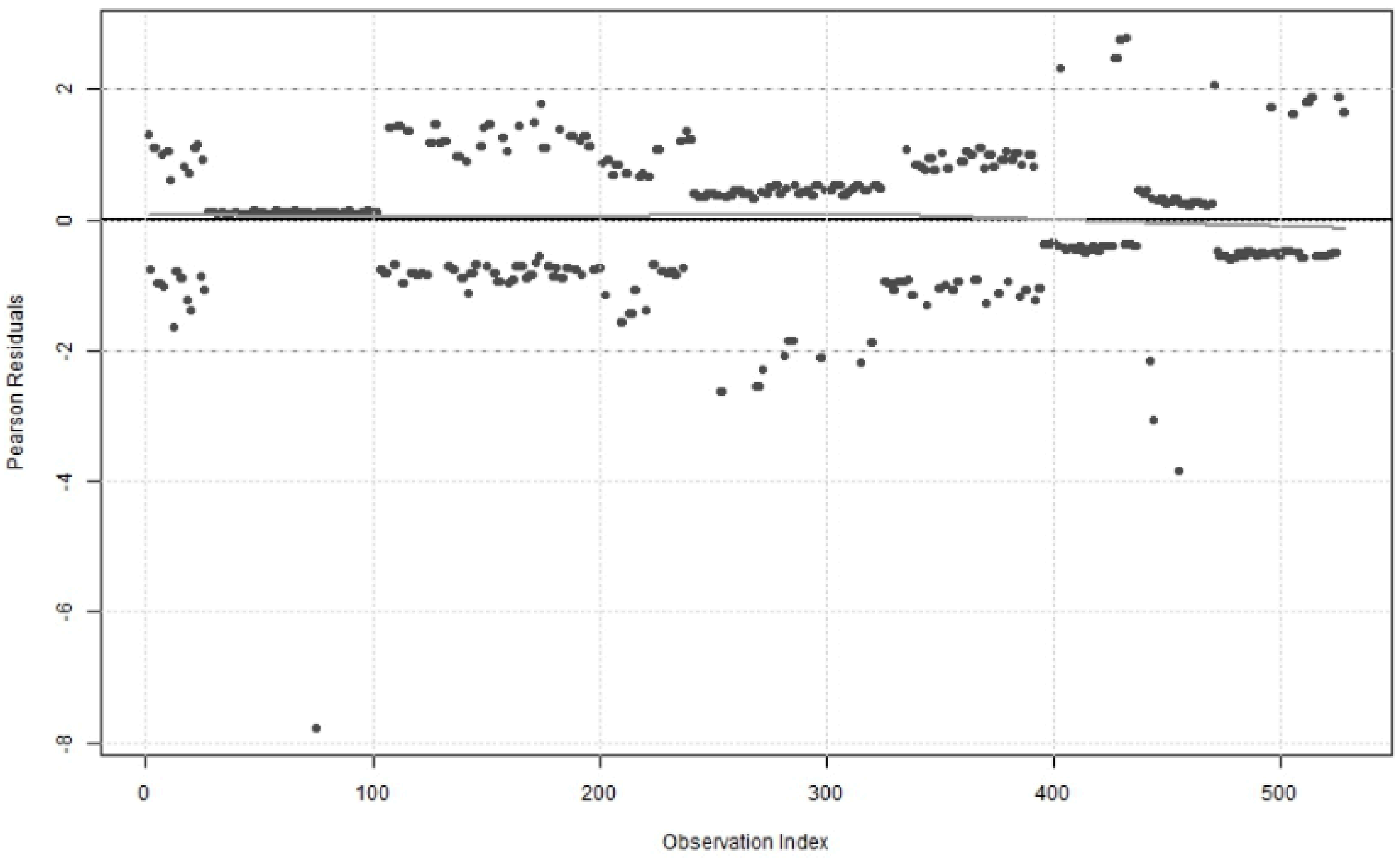
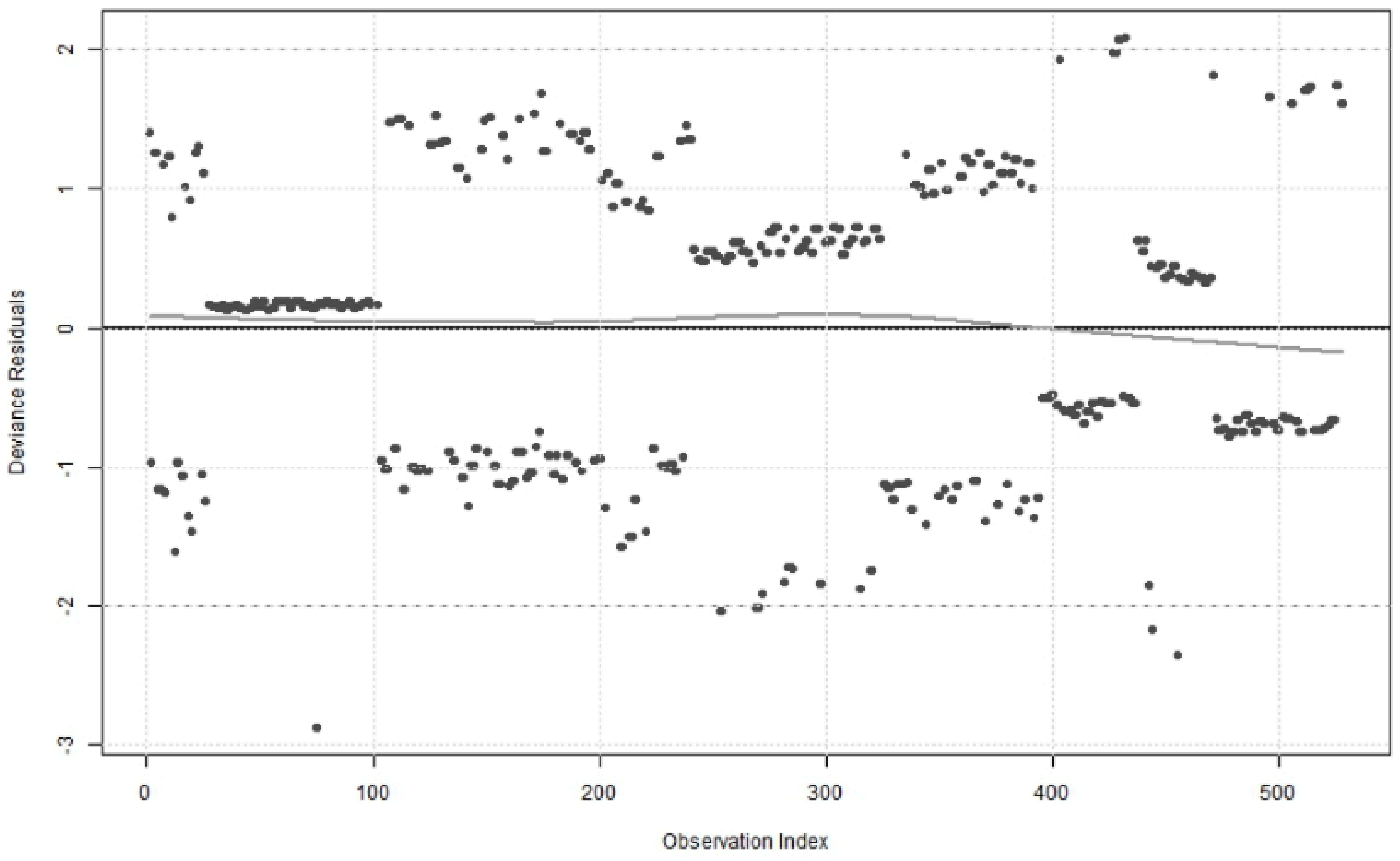


Figure 6: Deviance Residuals vs Observation Index



Pearson Residuals:

- Centered around zero with no clear pattern
- Most values within expected range (±2)
- A few extreme observations → potential outliers

Deviance Residuals:

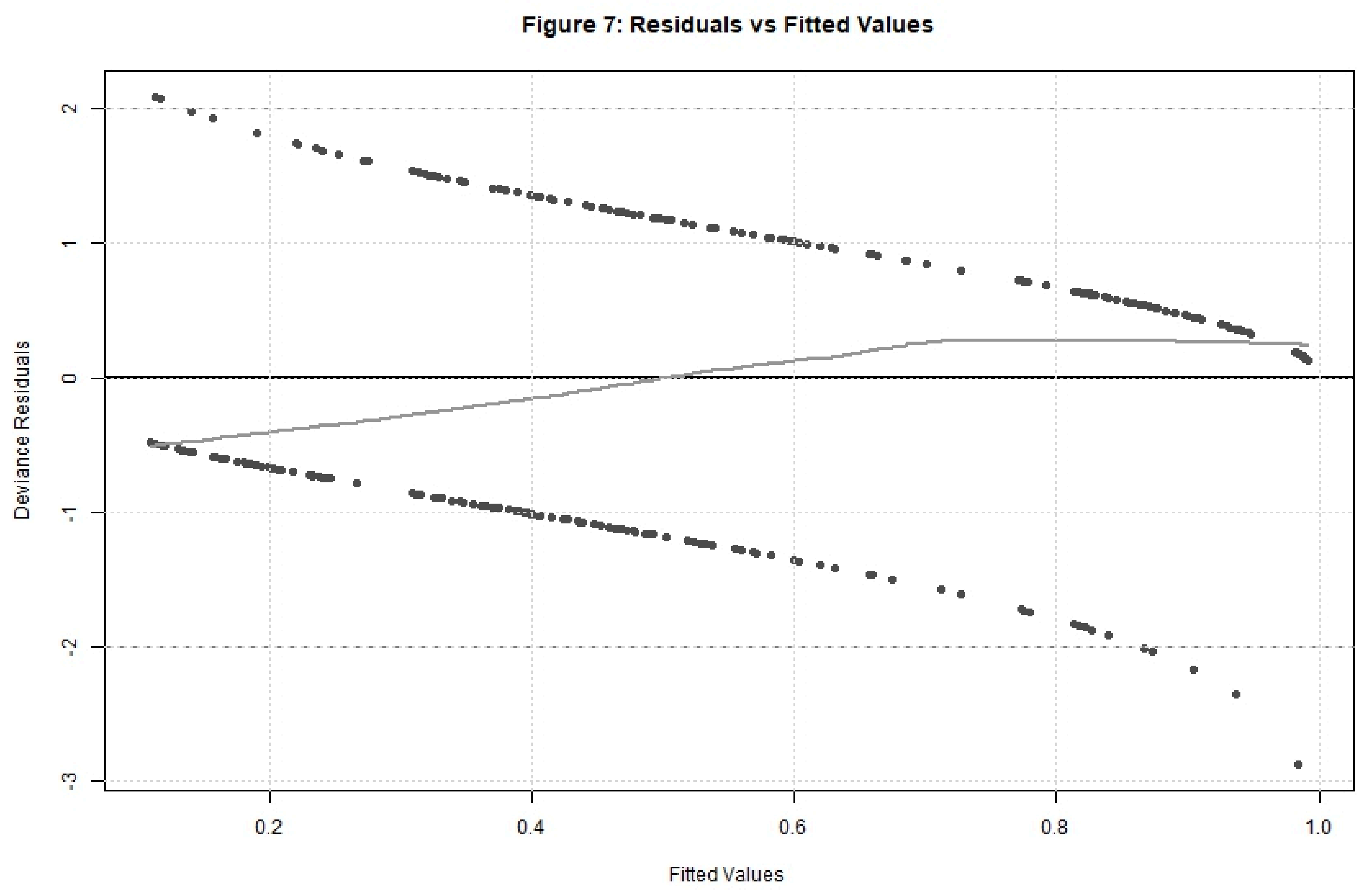
- Centered around zero with no systematic trend
- Clear two-band structure (expected for binary outcomes)
- Some large residuals, but limited impact

Multiple Logistic Regression Models

Model Diagnostics

Residuals vs Fitted Values:

- **Residuals form a clear two-band pattern**
  Expected for a binary outcome (logistic regression)
- **Points are centered around zero across fitted values**
  No major systematic bias
- **Smooth line is relatively flat with slight curvature**
  No strong evidence of model misspecification
- **A few larger residuals at high fitted values**
  Potential outliers, but limited impact
- **No evidence of systematic patterns, suggesting an appropriate model fit**



Model fits well across the range of predicted probabilities

Multiple Logistic Regression Models

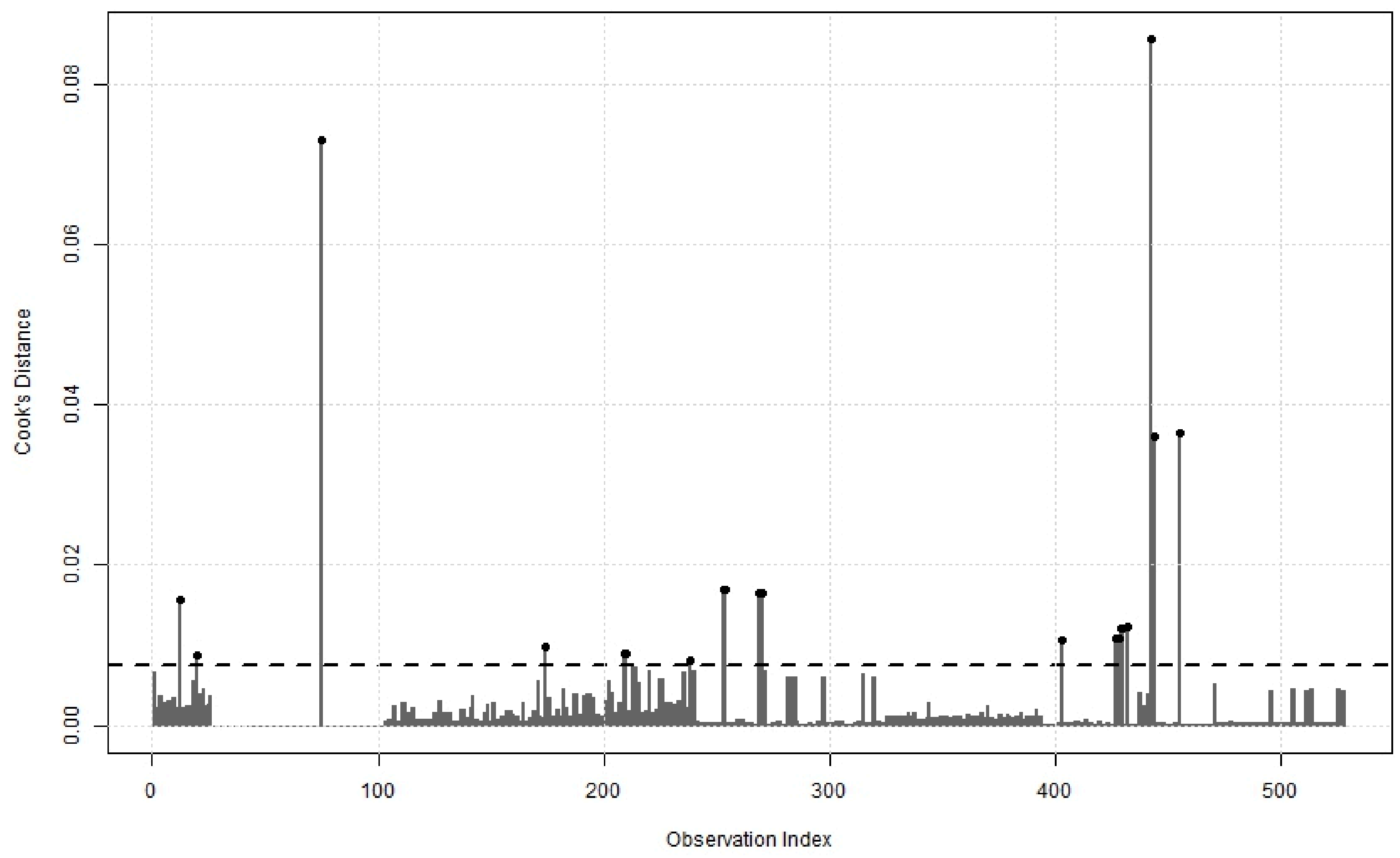
Influential Observations (Cook's Distance)

Table 21. Influential Observations Identified by Cook's Distance

Observations with Cook's Distance > 4/n

Observation	Cooks_Distance	Asthma_ED_Status	Annual_Weighted_AQI	State
442	0.0856	Not Elevated	35.98592	Rhode Island
75	0.0730	Not Elevated	29.77178	Florida
455	0.0365	Not Elevated	45.27397	South Carolina
444	0.0360	Not Elevated	29.18182	South Carolina
253	0.0169	Not Elevated	45.12363	New Jersey
254	0.0169	Not Elevated	45.12363	New Jersey
269	0.0166	Not Elevated	43.14404	New Jersey
270	0.0166	Not Elevated	43.14404	New Jersey
12	0.0157	Not Elevated	79.65753	Arizona
432	0.0124	Elevated	31.84932	Oregon
429	0.0121	Elevated	32.75623	Oregon
430	0.0121	Elevated	32.75623	Oregon
427	0.0108	Elevated	40.75342	Oregon
428	0.0108	Elevated	40.75342	Oregon
403	0.0106	Elevated	45.20548	Oregon
174	0.0099	Elevated	26.66667	Iowa
209	0.0090	Not Elevated	55.54795	Kansas
210	0.0090	Not Elevated	55.54795	Kansas
20	0.0087	Not Elevated	68.01370	Arizona
238	0.0082	Elevated	32.06997	Nebraska

Figure 8: Cook's Distance



Key Findings

- Influential points are isolated rather than widespread → no systematic instability
- Most observations have very low influence (near zero)
- A small number of observations exceed the 4/n threshold, with the largest ≈ 0.086 indicating moderate influence
- Influential observations are spread across multiple states (e.g., Rhode Island, Florida, South Carolina, New Jersey, Oregon)
- Include both outcome groups, though many are "Not Elevated" cases

Model results seem stable and not driven by individual observations

Multiple Logistic Regression Models

Chosen Model and Goodness of Fit

Model Form:

$$\log\left(\frac{P(\text{Asthma ED Status} = \text{Elevated})}{1 - P(\text{Asthma ED Status} = \text{Elevated})}\right) = \beta_0 + \beta_1(\text{Annual Weighted AQI}) + \beta_2(\text{State})$$

Expanded:

$$= \beta_0 + \beta_1(\text{AQI}) + \beta_2(\text{Florida}) + \beta_3(\text{Indiana}) + \cdots + \beta_k(\text{Wisconsin})$$

Key Findings

- Large reduction in deviance ($\Delta \approx 206$) → substantial improvement over null model
- Pearson $\chi^2 \approx$ df (512 vs 514), $p = 0.51$ → no evidence of lack of fit
- Hosmer–Lemeshow test non-significant ($p = 0.86$) → No evidence of lack of fit; model predictions align with observed outcomes

Model fits the data well with no evidence of lack of fit

Table 22. Model Fit and Goodness-of-Fit Statistics

Measure	Statistic	df	p-value
Model Deviance	513.51	NA	NA
Null Deviance	719.80	NA	NA
Deviance Difference (LRT)	206.29	NA	NA
Pearson Chi-Square	512.35	514	0.512
Hosmer–Lemeshow Test	3.98	8	0.859

Multiple Logistic Regression Models

Contribution of Annual Weighted AQI to Chosen Model Fit: Model Comparison

Likelihood Ratio Test

Table 23. Contribution of Annual Weighted AQI to Model Fit (Likelihood Ratio Test)

Comparison	Test Statistic (χ^2)	df	p-value
State-only model vs State + Annual Weighted AQI model	5.04	1	0.025

Key Findings

- Adding AQI improves model fit ($\chi^2 \approx 5.04$, $p = 0.025$)
- Reduction in deviance indicates better fit compared to state-only model

AQI provides significant additional explanatory value beyond state

Insights

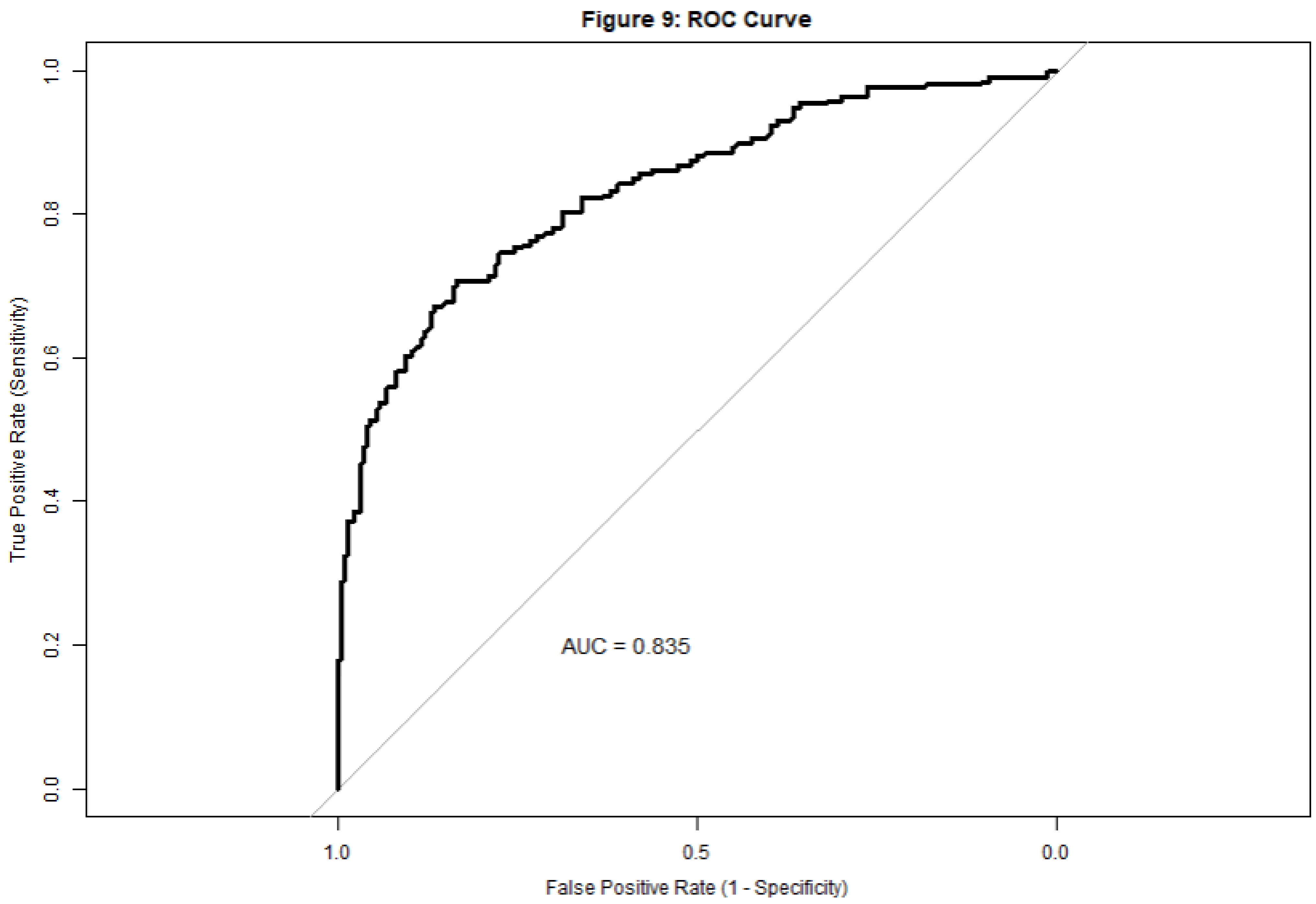
- AQI captures variation in asthma outcomes not explained by state alone
- Even after accounting for geographic differences, air pollution remains relevant
- Consistent with adjusted model where AQI remained a significant predictor

Multiple Logistic Regression Models

Receiver Operating Characteristic (ROC) Curve

Key Findings

- The ROC curve shows strong separation, with AUC = 0.835, indicating good discriminatory ability of the final model (Annual Weighted AQI + State)
- The steep rise at low false positive rates suggests the model can correctly identify many elevated asthma ED counties with relatively few false positives
- Performance improves steadily across thresholds, reflecting consistent contribution of both AQI and state-level effects in distinguishing outcomes



The model demonstrates good discrimination, effectively separating elevated from non-elevated asthma ED status

Multiple Logistic Regression Models

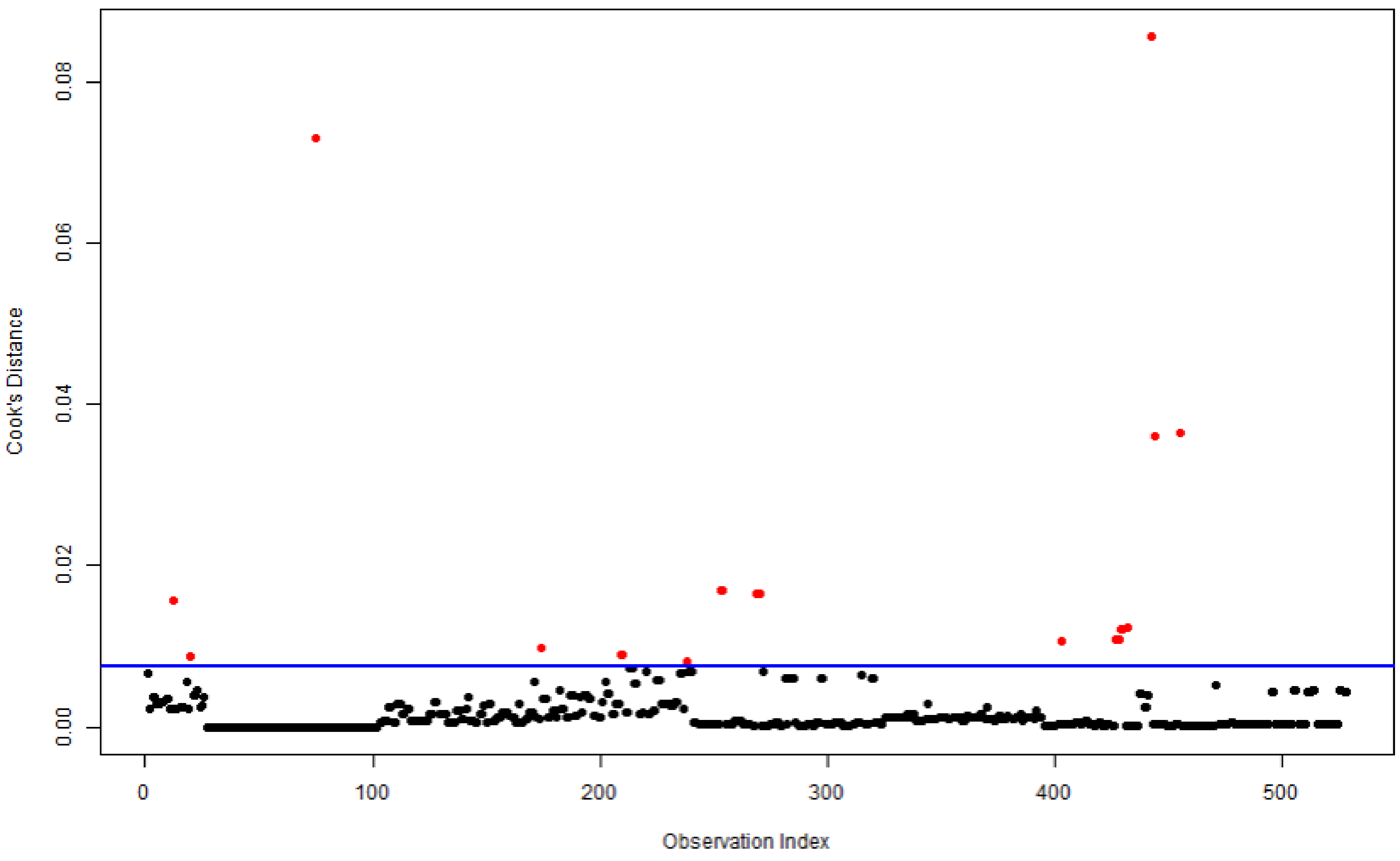
Sensitivity Analysis

Removing Influential Points

Purpose

- Evaluate the robustness of the AQI–asthma relationship
- Ensure conclusions are not dependent on extreme or unusual observations

Figure 10: Cook's Distance (Influential Points Highlighted)



Influential Points may reflect

- Extreme Weighted Annual AQI values
- Unusual asthma ED patterns

Important to verify whether these points are distorting the AQI effect

Multiple Logistic Regression Models

Comparison of Final Model With and Without Influential Observations

Table 24. Comparison of Final Model and Sensitivity Analysis Model (After Removing Influential Observations)

Model	Estimate	Std. Error	p_value	AIC	Deviance
Final Model	0.03	0.01	0.027	541.51	513.51
Cleaned Model	0.05	0.01	<0.001	434.49	406.49

Key Points

- **AQI effect becomes stronger after removing influential observations**
  Estimate increases (0.03 → 0.05)
- **Statistical significance improves ($p = 0.027$ → $p < 0.001$)**
  Model fit improves substantially
- **Lower AIC and deviance reflects improved model fit after removing influential observations**

Influential observations were attenuating the AQI effect

Insights

- After removing influential counties, AQI shows a clearer and stronger association with asthma ED status
- **Results suggest the relationship is robust and not driven by a few extreme observations**

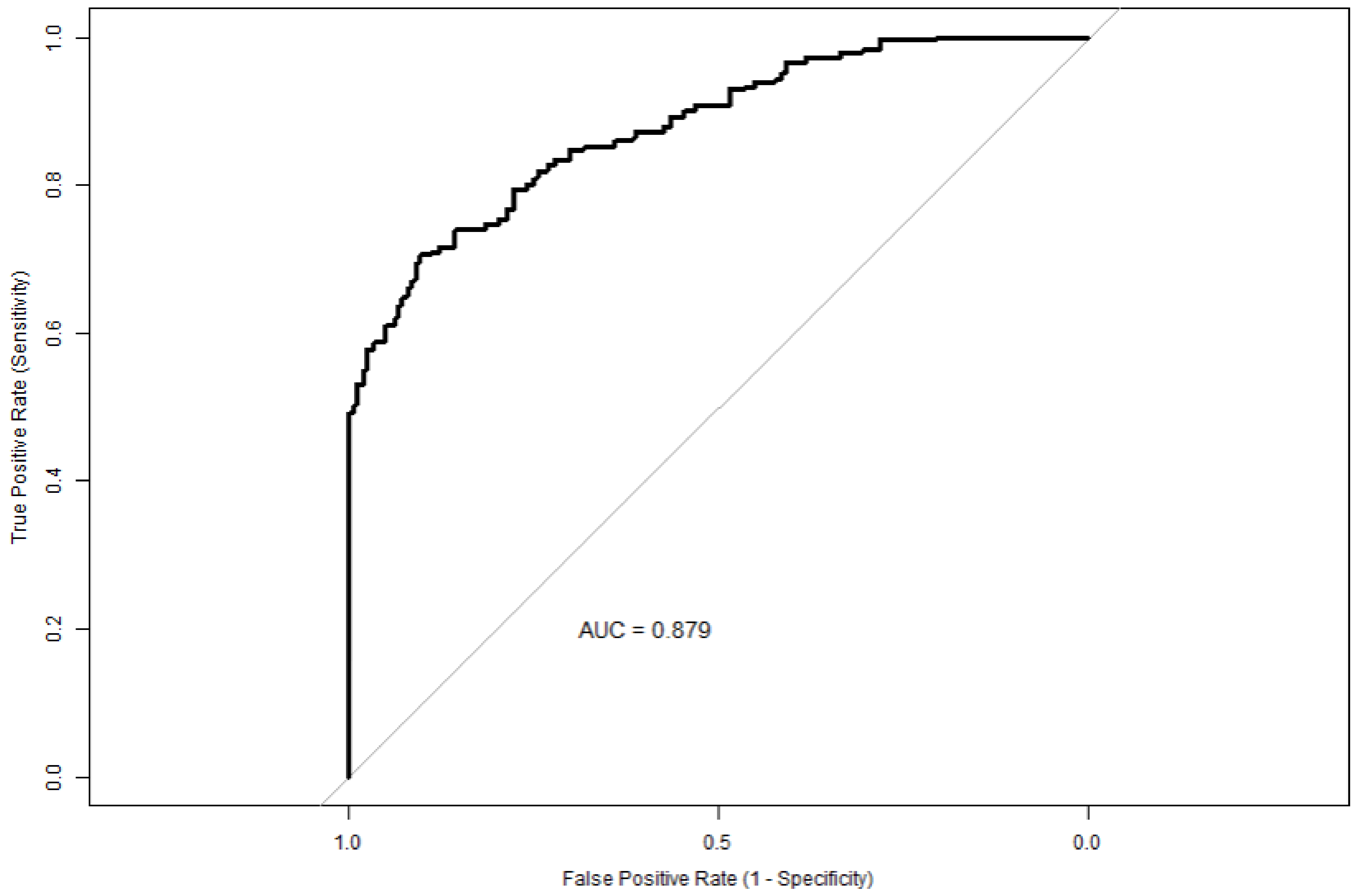
Multiple Logistic Regression Models

ROC Curve Comparison: Models With and Without Influential Observations

Key Findings

- AUC improves slightly (0.835 → 0.879)
- Both models demonstrate good discrimination
- Influential observations affect predictive performance, not conclusions
- The AQI–asthma relationship remains stable and consistent
- Improvement reflects a cleaner signal, not a new result

Figure 11. ROC Curve for Cleaned Model (After Removing Influential Observations)



The relationship is robust and not driven by a few extreme observations

Multiple Logistic Regression Models

Model Testing and Validation

Real differences in asthma burden across states that are not explained by AQI

State-Level Asthma Burden (Adjusted for AQI)

Estimated state effects from the cleaned logistic regression model after adjusting for Annual Weighted AQI

Table 25. State-Level Effects on Asthma ED Status (Adjusted for Annual Weighted AQI)

States ranked by log-odds (lowest to highest relative to reference state)

State	Log-Odds	Std. Error	p-value
Oregon	-19.760	1769.105	0.991
Wisconsin	-1.637	0.539	0.002
Iowa	-1.212	0.592	0.041
Indiana	-0.679	0.501	0.176
Nebraska	-0.637	0.668	0.340
North Carolina	0.100	0.493	0.840
Kansas	0.688	0.658	0.296
New York	1.595	0.561	0.005
New Jersey	19.335	2087.261	0.993
South Carolina	19.435	2074.495	0.993
Rhode Island	19.515	4780.413	0.997
Florida	19.603	1218.231	0.987

Key Findings

- Lowest adjusted burden:
  Wisconsin (p = 0.002)
  Iowa (p = 0.041)
- Highest adjusted burden:
  New York (p = 0.005)

Interpreting State Effects (E.g. Wisconsin)

- Log-Odds = -1.637
- p-value = 0.02 (statistically significant)

At the same AQI level, counties in Wisconsin have approximately 81% lower odds of elevated asthma ED visits compared to the reference state (OR ≈ 0.19).

Geographic differences in asthma burden persist even after adjusting for AQI

Multiple Logistic Regression Models

Model Testing and Validation

Predicted Asthma ED Risk Across AQI Levels (State Held Constant)

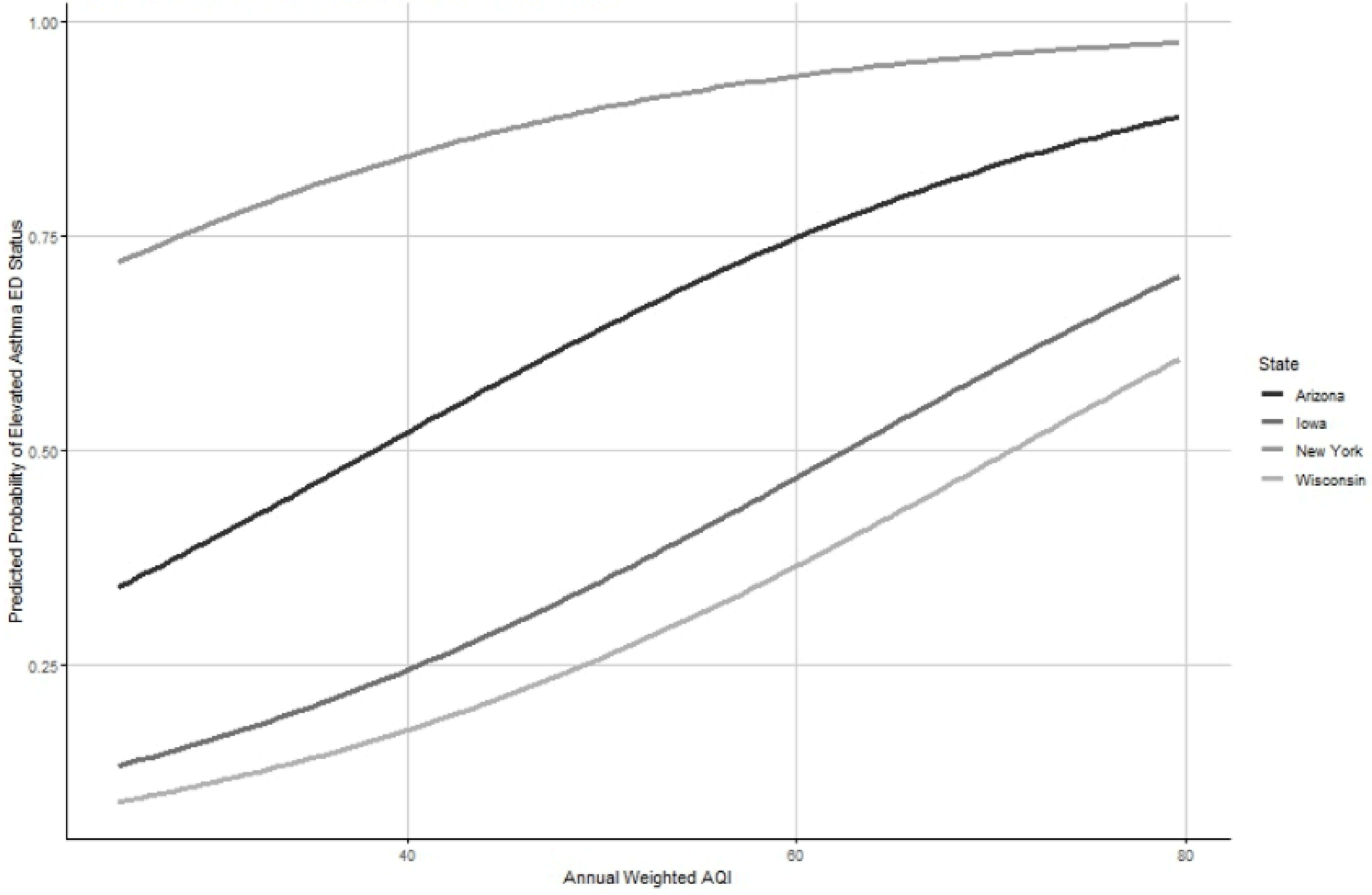
Key Findings

- Predicted probability of elevated asthma ED status increases with AQI in all states
- Indicates a **consistent positive** relationship between air pollution and asthma risk

State Comparisons

- New York → consistently highest predicted risk
- Wisconsin → consistently lowest predicted risk
- Arizona & Iowa → intermediate risk levels

Figure 12: Effect of AQI on Asthma ED Risk Across States



- Curves are upward sloping → higher AQI → higher risk
- Curves are roughly parallel → AQI effect is similar across states
- Curves are vertically separated → states differ in baseline risk

Air pollution increases asthma risk everywhere, but state-level factors determine how high that risk is

Multiple Logistic Regression Models

DISCUSSION

Why the AQI Relationship Appeared Negative Initially

Key Findings

- Early model suggested a negative AQI–asthma relationship

This was driven by:

- Influential observations
- Between-state differences (confounding)

What Changed

- Removed influential observations
- Adjusted properly for state-level variation

The negative relationship was an artifact; the true relationship is positive

CONCLUSION

KEY FINDINGS

- AQI is significantly associated with asthma ED risk
  Higher AQI → higher probability of elevated asthma ED visits
- State-level differences persist after adjustment
  Asthma burden varies even at similar AQI levels

MODEL INSIGHTS

- Final model (AQI + State) provided the best fit
- No strong interaction → AQI effect is consistent across states
- Sensitivity analysis confirmed results are robust

PREDICTIVE PERFORMANCE

- Good discrimination (AUC improved from 0.835 → 0.879)
- Removing influential observations refined model performance

PUBLIC HEALTH IMPLICATIONS

- Reducing AQI may lower asthma-related ED visits
- However, state-specific factors must also be addressed

AQI matters—but it is not the whole story

References

1. NIEHS (Air Pollution & Health Effects)
National Institute of Environmental Health Sciences. (n.d.). Air pollution.
https://www.niehs.nih.gov/health/topics/agents/air-pollution

2. CDC (Air Pollution & Health Effects)
Centers for Disease Control and Prevention. (2024). Air pollution and health.
https://www.cdc.gov/climate-health/php/effects/air-pollution.html

3. EPA (Health Effects of Air Pollution)
U.S. Environmental Protection Agency. (n.d.). Research on health effects of air pollution.
https://www.epa.gov/air-research/research-health-effects-air-pollution

4. CDC (Asthma Data / Tracking Network)
Centers for Disease Control and Prevention. (2023). Asthma data and surveillance.
https://www.cdc.gov/environmental-health-tracking/php/data-research/asthma.html

5. AAFA (Asthma Facts)
Asthma and Allergy Foundation of America. (n.d.). Asthma facts and figures.
https://aafa.org/asthma/asthma-facts/

6. Agresti, A. (2018). Statistical Methods for the Social Sciences (5th ed.). Pearson.

R Packages Used:

readxl, dplyr, ggplot2, gtsummary, gt, vcd, tidyr, scales, DescTools, broom, MASS, ResourceSelection, pROC

Thank You